

Digital Investigation

Constructed by Ademide Ajiboye (Job Simulation)

Executive Summary

As part of a simulated digital investigation exercise, network traffic captured from a flagged laptop on the BNS Cyberlab Limited network was analyzed to identify downloaded files, viewed images, and other artifacts of interest. Using Wireshark, HTTP traffic was filtered and TCP streams were reconstructed and carved at the hex level to extract file content, which was then verified using a hex editor and scanned via VirusTotal for malicious indicators.

The investigation identified four files transferred over HTTP: a company-branded image (BNS logo/graphic, both **PNG** and **JPG**), a **PDF** containing an image with accompanying description and reference information on network traffic analysis, and a **ZIP** archive containing internal BNS financial reports, including a balance sheet and cash flow statement. VirusTotal analysis returned no malicious detections across any of the extracted files, and no malware indicators or command-and-control activity were identified.

It is recommended that BNS Cyberlab Limited (1) confirm whether the financial report transfer was authorized and legitimate, (2) enforce encrypted channels (HTTPS) for any transfer of sensitive internal documents, and (3) review access controls around financial reporting data to prevent unauthorized or unencrypted distribution going forward.

Tools used:

Wireshark — packet analysis, protocol dissection, stream reconstruction
Hex editor — raw file carving/verification, file signature checks
VirusTotal — malware/URL reputation scanning

Step 1 - Traffic Filtering:

Applied Wireshark display filter “http” to isolate HTTP requests/responses from the full capture, narrowing focus to web traffic likely to contain file transfers or downloaded content.

bns_captured_activity.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

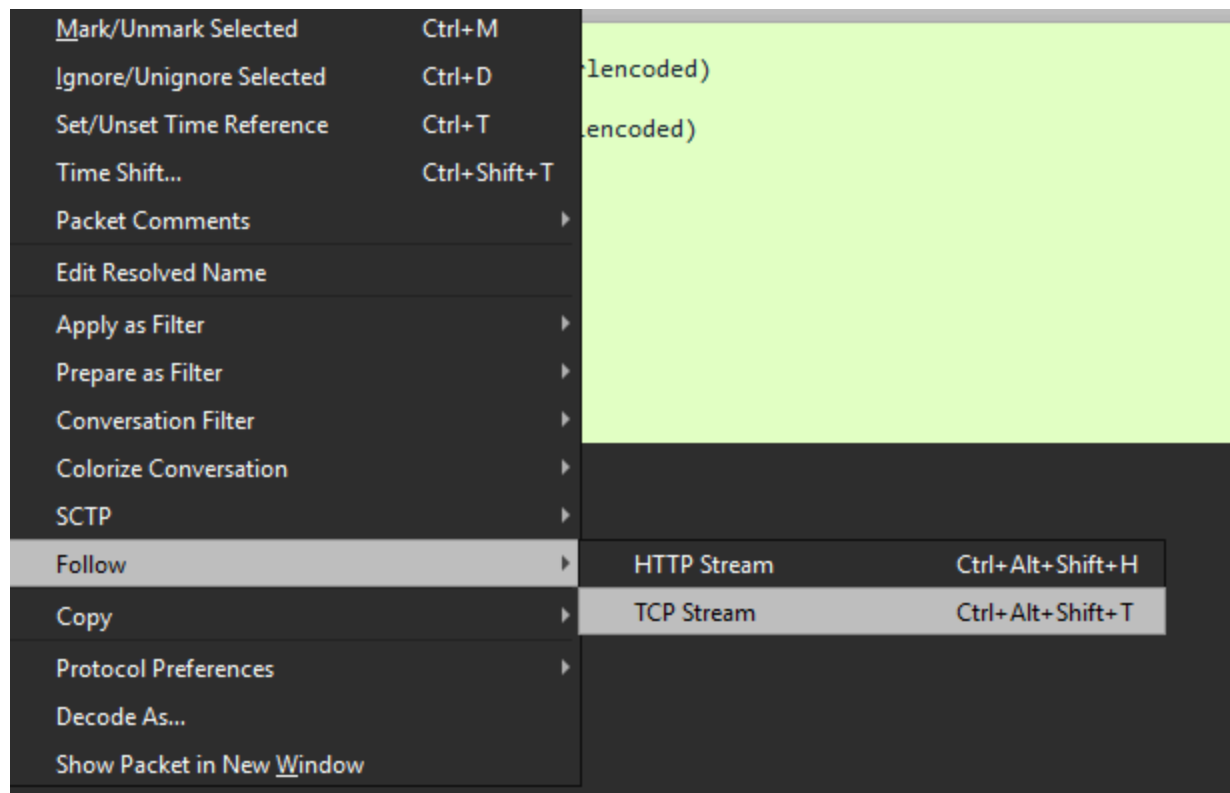
http

No.	Time	Source	Destination	Protocol	Length	Info
4	0.000000	192.168.0.2	192.168.0.115	HTTP	173	GET /BNS02.png HTTP/1.1
20	0.000000	192.168.0.115	192.168.0.2	HTTP	994	HTTP/1.1 200 OK (PNG)
26	0.000000	192.168.0.2	192.168.0.115	HTTP	174	GET /BNS01.jpg HTTP/1.1
153	0.046902	192.168.0.115	192.168.0.2	HTTP	41	HTTP/1.1 200 OK (JPEG JFIF image)
159	-2441.910682	192.168.0.2	192.168.0.115	HTTP	190	POST /login HTTP/1.1 (application/x-www-form-urlencoded)
160	-2439.910682	192.168.0.115	192.168.0.2	HTTP	111	HTTP/1.1 200 OK (text/plain)
161	-2431.910682	192.168.0.2	192.168.0.115	HTTP	227	POST /chat HTTP/1.1 (application/x-www-form-urlencoded)
162	-2427.910682	192.168.0.115	192.168.0.2	HTTP	106	HTTP/1.1 200 OK (text/plain)
166	0.062508	192.168.0.2	192.168.0.115	HTTP	179	GET /BNS03.pdf HTTP/1.1
239	0.078123	192.168.0.115	192.168.0.2	HTTP	1260	HTTP/1.1 200 OK (application/pdf)
245	-2986.910682	192.168.0.2	116.193.111.166	HTTP	163	GET / HTTP/1.1
246	-369.910682	192.168.0.2	124.58.116.245	HTTP	164	GET / HTTP/1.1
247	-2218.910682	192.168.0.2	72.220.211.91	HTTP	163	GET / HTTP/1.1
248	-3394.910682	192.168.0.2	139.209.142.46	HTTP	164	GET / HTTP/1.1
249	-3499.910682	192.168.0.2	24.215.60.239	HTTP	163	GET / HTTP/1.1
253	0.078123	192.168.0.2	192.168.0.115	HTTP	179	GET /BNS04.zip HTTP/1.1
416	0.130744	192.168.0.115	192.168.0.2	HTTP	518	HTTP/1.1 200 OK (application/zip)

Step 2 — Stream Identification

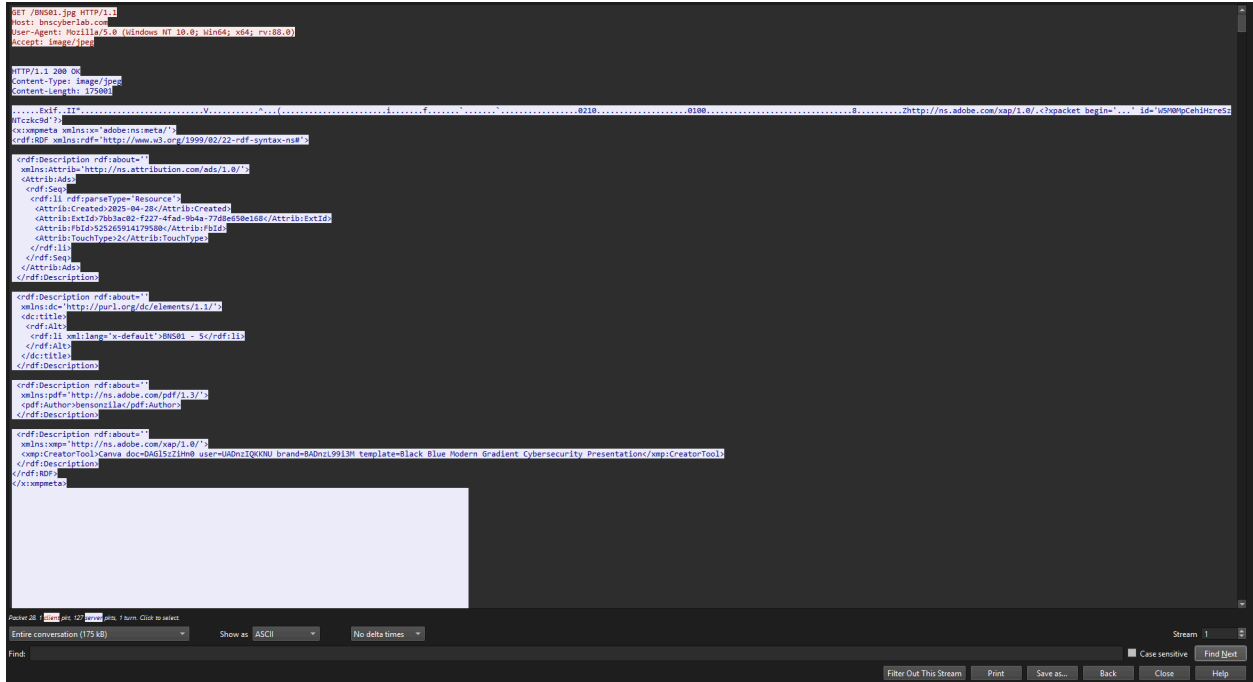
Reviewed HTTP GET/POST responses for Content-Type headers indicating target file types (BNS01/jpeg, BNS02/png, BNS03/pdf, BNS04/zip). Identified candidate packets and right-clicked → Follow → TCP Stream to reconstruct the full conversation.

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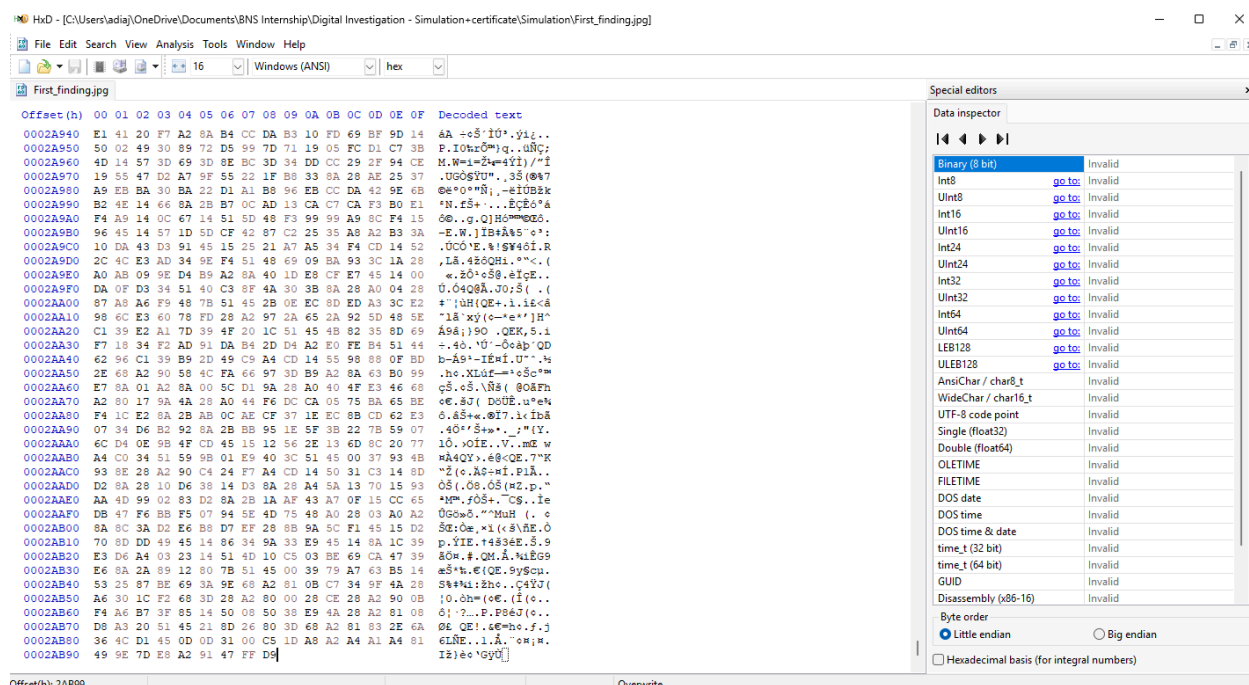
Step 3 - Raw extraction:

In the Follow TCP Stream window, changed the "Show data as" dropdown from the default ASCII view to Raw, isolating the pure byte stream (headers + payload) without Wireshark's text rendering interfering with binary content.



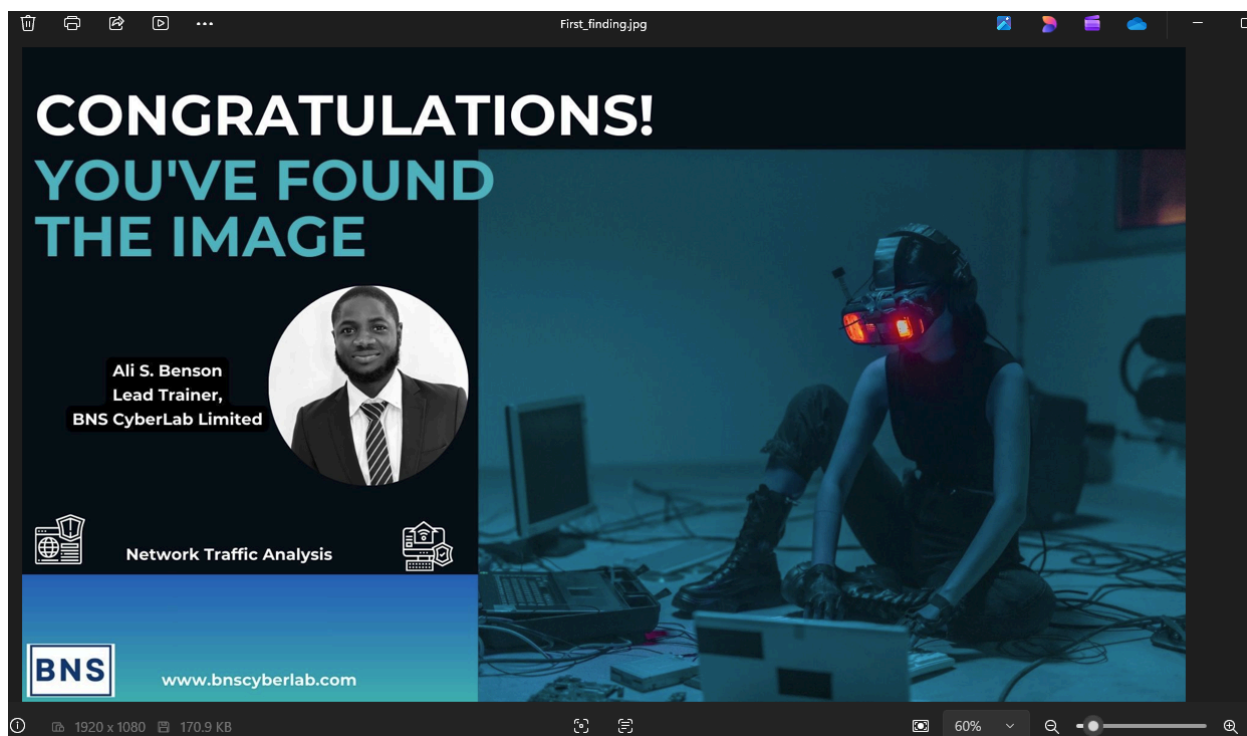
Step 4 – Hex-Leve Carving

Copied the raw stream data into a hex editor. Located the HTTP response body boundary/end of headers, marked by `\r\n\r\n` and identified the file's magic bytes/signature (eg. `FF D8 FF` for JPEG) to confirm file type and isolate the payload from HTTP header noise.



Step 5 — File Reconstruction & Verification

Saved the carved binary data as a standalone file (e.g. First_finding.jpg), then verified integrity by opening it and/or checking the file signature matched the expected format.



Step 6 — Repeat for Remaining File Types

Repeated Steps 1–5 for PNG, PDF, and ZIP artifacts, adjusting the expected magic bytes for each:

JPG - FF D8 FF

PNG - 89 50 4E 47 0D 0A 1A 0A

PDF - 25 50 44 46

ZIP - 50 4B 03 04



Third_finding.pdf

1 / 184%+

1

NETWORK TRAFFIC ANALYSIS

Network Traffic Analysis is the process of capturing, inspecting, and interpreting data as it moves across a network to understand what's happening inside that network.

why is network traffic analysis important?

Network traffic analysis is important because it helps detect cyber threats, monitor abnormal activities, investigate security incidents, protect sensitive data, and optimize network performance. By understanding and inspecting traffic patterns, organizations can respond quickly to attacks, prevent data breaches, ensure compliance with regulations, and maintain a secure, efficient network environment.

Congratulations! PDF file downloaded

20%

Fourth_finding.zip (evaluation copy)

File Commands Tools Favorites Options Help

Add

Extract To

Test

View

Delete

Find

Wizard

Info

VirusScan

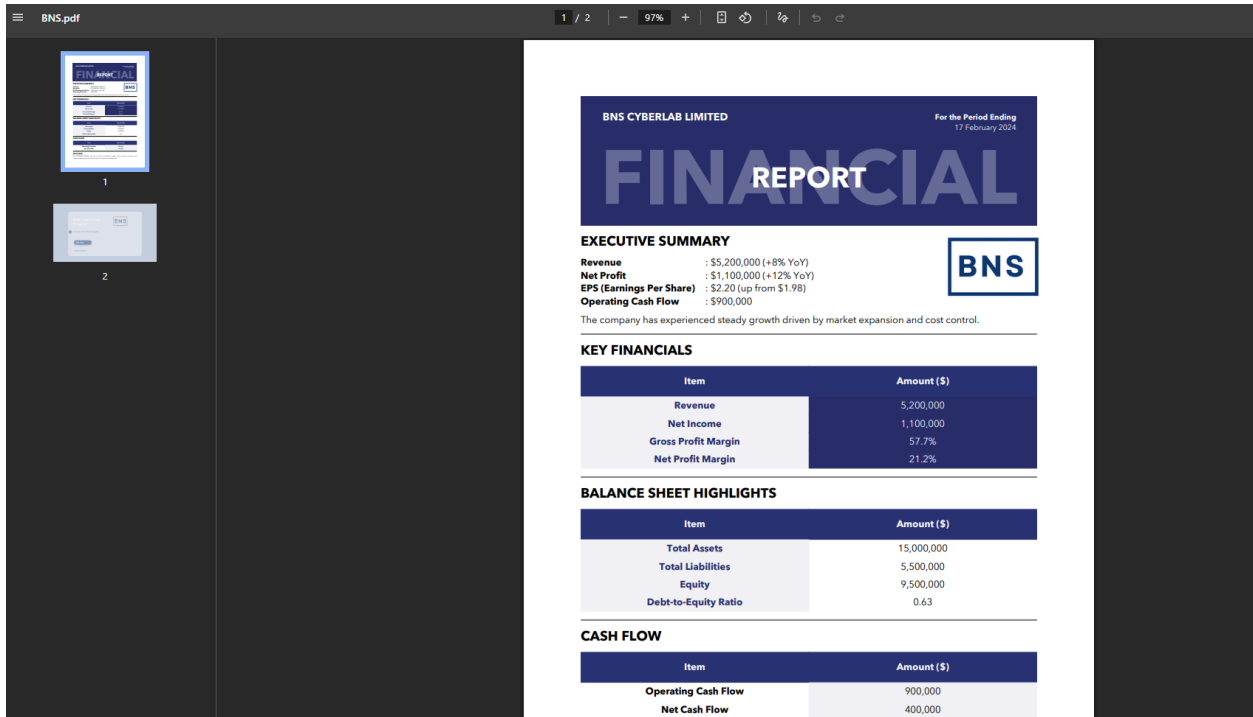
Comment

SFX



Fourth_finding.zip - ZIP archive, unpacked size 238,852 bytes

Name	Size	Packed	Type	Modified	CRC32
..			File folder		
 BNS.pdf	234,931	221,894	Brave PDF Docum...	29/04/2025 08:...	224F1D63
 bns33.jpeg	3,921	3,776	JPEG File	29/04/2025 07:...	2A9EBFC5



VirusTotal scans:

